

Decomposition Formula Used in the Slide Figure

Setup

For each year t , the model estimates:

- ω_θ : population share of type θ (fixed at trend values in Phase 3)
- $u_\theta(t)$: ergodic unemployment *mass* of type θ (i.e., the fraction of type θ that is unemployed in the ergodic distribution of its transition matrix in year t)

Step 1: Aggregate unemployment mass

The aggregate unemployment mass (fraction of the total population that is unemployed) is:

$$U_{\text{agg}}(t) = \sum_{\theta=1}^5 \omega_\theta \cdot u_\theta(t) \quad (1)$$

Step 2: Change in aggregate unemployment

For a recession with peak year t_0 and trough year t_1 :

$$\Delta U_{\text{agg}} = U_{\text{agg}}(t_1) - U_{\text{agg}}(t_0) = \sum_{\theta=1}^5 \omega_\theta \cdot [u_\theta(t_1) - u_\theta(t_0)] = \sum_{\theta=1}^5 \omega_\theta \cdot \Delta u_\theta \quad (2)$$

Step 3: Type θ 's share of the aggregate increase

$$\boxed{\text{Share}_\theta = \frac{\omega_\theta \cdot \Delta u_\theta}{\sum_{\theta'=1}^5 \omega_{\theta'} \cdot \Delta u_{\theta'}} \times 100\%} \quad (3)$$

By construction, $\sum_{\theta=1}^5 \text{Share}_\theta = 100\%$.

Since $\omega_{\text{All-N}}$ types have $u_{\text{All-N}}(t) = 0$ for all t , All-N always contributes zero.

Step 4: Renormalization to mover types only (used in slide figure)

The slide figure shows only the three mover types (High-N, High-U, High-E), renormalized:

$$\text{Share}_\theta^{\text{mover}} = \frac{\text{Share}_\theta}{\text{Share}_{\text{High-N}} + \text{Share}_{\text{High-U}} + \text{Share}_{\text{High-E}}} \times 100\%, \quad \theta \in \{\text{High-N}, \text{High-U}, \text{High-E}\} \quad (4)$$

Difference from the paper's formula

The **paper** uses the unemployment *rate* decomposition:

$$c_{\theta}(t) = \frac{\omega_{\theta} \cdot u_{\theta}(t)}{\underbrace{\sum_{\theta'} \omega_{\theta'} \cdot [u_{\theta'}(t) + e_{\theta'}(t)]}_{LF_{\text{agg}}(t)}}, \quad \text{Share}_{\theta}^{\text{paper}} = \frac{\Delta c_{\theta}}{\Delta UR} \quad (5)$$

The difference: the paper divides by $LF_{\text{agg}}(t)$ which *changes between peak and trough*. This introduces cross-terms that can make All-E's contribution large and negative (up to -79% in 2020), because All-E's $u_{\text{All-E}}$ fluctuates slightly and is multiplied by $\omega_{\text{All-E}} \approx 0.60$.

The slide formula (eq. 3 above) avoids this by working with unemployment *mass* directly, without dividing by a time-varying labor force.